# Does the Great Mortgaging Crowd Out Small Firms?

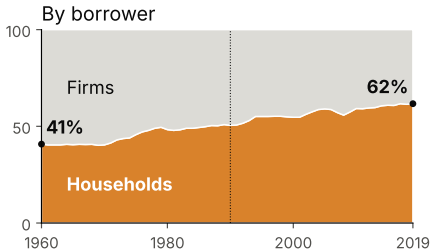
## Mortgage lending, bank capital and the size distribution of firms

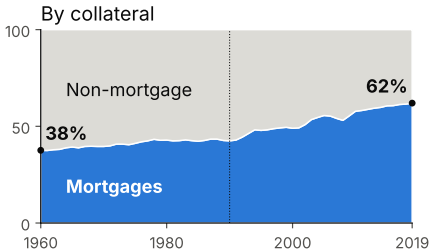
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# Motivation: the Great Mortgaging




Average of 11 countries, 1960–2019: firms **59%** → **38%** of bank credit, mortgages **38%** → **62%**. Over the century: mortgages about 30% around 1900, about 60% today (JST data; Jordà, Schularick & Taylor 2016)

**Question:** did this shift crowd out lending to **small, bank-dependent firms**, while large firms turned to bond markets?

# A tension in the literature

**US counties: Chakraborty et al. (2019)**

- 1988–2006, banks in strong housing markets
- **More** mortgages, **less** commercial lending
- Their borrowers invest less, especially constrained firms and banks

**Spain: Martín, Moral-Benito, Schmitz (2021)**

- Housing boom **first reduces** credit to non-housing firms
- **Later stimulates** it: the boom raises bank net worth
- Crowding-out large but **temporary**, absorbed by the end of the boom

**So is crowding-out only a short-run effect?** The aggregate effect can fade while its *composition* does not: who loses credit, and do they come back?

# My idea: firm size

**Bank with limited capital**

A euro of capital must earn the same in both uses:

$$R_t^f - R_t^d = \frac{\omega^f}{\omega^m}\left(R_t^m - R_t^d\right)$$

Mortgages pay more $\Rightarrow$ the bank asks more from firms.

$R^f, R^m$: firm and mortgage rates; $R^d$: funding cost;
$\omega^f, \omega^m$: capital charged per euro lent (Bâle I: 1 and 0.5).

**Short run (capital fixed)**

Firms pay a higher rate: small firms invest less or exit. Large firms escape through bonds.

**Long run (capital adjusts)**

Crowding-out in levels fades (as in Spain), but firms stay structurally disadvantaged, and small firms that exited may not come back.

# Usable data

| Source | What it gives | Coverage |
|---|---|---|
| County Business Patterns | Establishments and jobs by size class and sector | County, yearly |
| CRA | Small-business loans by bank and county | From 1996 |
| HMDA | Mortgage lending by bank | From the 1990s |
| FDIC Summary of Deposits | Where each bank has branches | From 1994 |
| Call Reports | Bank balance sheets, capital | Quarterly, long run |
| FHFA + Saiz (2010) | House prices + housing supply elasticity | County / metro |
| JST Macrohistory | Mortgage and non-mortgage credit, 17 countries | Since 1870 |

All public, already collected.

# Identification: the Saiz instrument

**Push banks toward mortgages for reasons unrelated to local firms**

Housing booms in the *other* counties where a bank has branches, predicted by how hard it is to build there (Saiz 2010):

$$\text{Expo}_c = \sum_{\text{banks } b} \underbrace{s_{b,c}}_{\text{share in } c} \sum_{j \neq c} \underbrace{w_{b,j}\, \widehat{\Delta HPI}_j}_{\text{booms elsewhere}}$$

**Step 1: banks**

More mortgages, fewer small-business loans? Stronger when capital is low?

**Step 2: firms, 1 to 15 years**

Small vs large establishments in the same county. Does the gap close, as in Spain, or persist?

# Backup: the bank's problem

$$\max_{L_t^m, L_t^f, S_t, D_t} \; R_t^m L_t^m + R_t^f L_t^f + R_t^s S_t - R_t^d D_t \quad \text{s.t.} \quad L_t^m + L_t^f + S_t = N_t + D_t, \quad \omega^m L_t^m + \omega^f L_t^f \leq \phi N_t$$

$$\frac{R_t^m - R_t^d}{\omega^m} = \lambda_t = \frac{R_t^f - R_t^d}{\omega^f} \qquad dL_t^f = -\frac{\omega^m}{\omega^f} dL_t^m \;\; (N_t \text{ fixed})$$

**Long run** ($N_t$ chosen, cost $R_t^e$):

$$\lambda_t = \frac{R_t^e - R_t^d}{\phi} \quad \Rightarrow \quad R_t^f = R_t^d + \frac{\omega^f}{\phi}\left(R_t^e - R_t^d\right)$$

Extended version adds default risk, collateral value (house prices) and information costs.

# Backup: step 2 regression

$$Y_{c,s,k,t+h} - Y_{c,s,k,t-1} = \beta_h \, \text{Expo}_{c,t} \times \text{Small}_k + \theta_h \, \Delta HPI_{c,t} \times \text{Small}_k + \delta_{c,t} + \mu_{s,k,t} + \varepsilon$$

- $c$ county, $s$ sector, $k$ size class, $h = 1, \ldots, 15$ years
- Prediction: $\beta_h < 0$; persistence tells short run from long run
- Local house prices control for the home-collateral channel (Adelino, Schoar & Severino 2013)
- Robustness: tradable sectors only (manufacturing)

# Backup: larger project

- **Bâle I (1988)**: mortgages charged half the capital of firm loans, a historical shock to $\omega^m/\omega^f$
- Back to the long-run trend across JST countries: land vs productive capital
- Model extension: rising cost of capital, or exit with entry costs, to make the effect persist